ICSForge

OT/ICS Security Coverage Validation Platform

ICSForge™ is an open-source OT/ICS Security Coverage Validation Platform.

It aims to solve a real problem “How do I safely validate OT security countermeasures from functionality, visibility and detection coverage perspective by using industrial-looking traffic?”

The goal is simple: make it easier to answer “Are we actually seeing, detecting, and controlling what we think we are?”

Motivation and Problem Statement

If we take a look to the recent Dragos 2026 Year in Review report (https://www.dragos.com/blog/dragos-2026-ot-cybersecurity-year-in-review);
Based on previous incidents that have been studied on that report:
- 81% of reports showed poor IT/OT segmentation, 53% had exposed internet-facing assets.
- 46% of architecture reviews and 88% of tabletops showed deficient detection.

Modern OT environments increasingly deploy security controls such as:
- OT-aware firewalls and segmentation gateways
- Network Security Monitoring (NSM) sensors
- IDS/NDR platforms
- SIEM pipelines consuming OT telemetry

Despite this investment, many organizations cannot confidently answer basic operational questions:
- What OT/ICS traffic is actually visible to our sensors?
- Are detection rules triggering when they should?
- Are segmentation and firewall policies enforced as designed?
- Can SOC analysts distinguish expected OT behavior from suspicious or policy-violating activity?

ICSForge was created to address this gap by enabling repeatable, protocol-aware validation of OT network visibility and detection coverage using controlled, inspectable traffic generation.

Detailed project outline is here: https://www.icsforge.nl

Source code can be found here: https://github.com/ICSForge/ICSForge

ICSForge Demo Lab (Flying IT/OT Lab)

ICSForge Demo Lab is a cabin-size mobile IT/OT lab and it has three zones:
IT Zone, DMZ and OT Zone.

IT distribution switch and ICSForge Sender (with display) are in IT Zone.
IT/OT firewall is separating IT and OT zone, and network security monitoring sensor is in DMZ.
SCADA server, touchscreen HMI, CO_2 Tank, OT distribution switch and ICSForge Receiver (with display) are in OT Zone.

ICSFORGE PROJECT DETAILS

ICSForge™

OT/ICS Cybersecurity Coverage Validation Platform

tests passing | python 3.10+ | License GPLv3 | version 0.59.4

ICSForge™ is an open-source **OT/ICS security coverage validation platform** designed to help defenders, SOC teams, and OT security engineers validate detection, visibility, and readiness against real-world industrial attack techniques.

ICSForge focuses on what can actually be observed on the network and generates realistic OT traffic and PCAPs in **10 industrial protocols (Modbus/TCP, DNP3, S7comm, IEC-104, OPC UA, EtherNet/IP, BACnet/IP, MQTT, IEC 61850 GOOSE, PROFINET DCP)** which are aligned with **67 of 83 unique techniques in MITRE ATT&CK for ICS v18 (80.7% coverage)** — without exploiting real systems or causing unsafe process impact — to help asset owners and defenders assess the quality of existing security countermeasures such as firewalls, OT NSM sensors and ACLs and identify hidden gaps.

ICSForge is developed with a **safe-by-design** approach, operating within a **Sender-Receiver architecture** and interacting only with the designated sender and receiver, without touching other OT devices.

> Most ICS security tools promise coverage - ICSForge lets you **prove it**.

Key Numbers (v0.59.4)

Metric	Value
Protocols	10 industrial protocols (Modbus/TCP, DNP3, S7comm, IEC-104, OPC UA, EtherNet/IP, BACnet/IP, MQTT, IEC 61850 GOOSE, PROFINET DCP)
Runnable Scenarios	536 standalone + 11 named attack chains = 547 total
ATT&CK for ICS Techniques Implemented	67 unique technique IDs in scenario steps
ATT&CK for ICS Matrix	83 unique technique IDs (94 total entries — 11 appear under multiple tactics)
ATT&CK for ICS Matrix Coverage	80.7% (67 out of 83 technique) — 35 of 67 techniques at 10/10 protocols
Detection Rules	Auto-generated Suricata + Sigma rules per scenario

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Architecture

```
┌──────────────────────┐                  ┌──────────────────────┐
│   ICSForge Sender    │  TCP/ UDP / L2   │  ICSForge Receiver   │
│                      │ ->->->->->->->-> │                      │
│ • Scenario engine    │                  │ • Traffic sink       │
│ • 10 protocol builders│                 │ • Marker correlation │
│ • PCAP generation    │                  │ • Receipt logging    │
│ • Campaign playbooks │   <-- callback   │ • Coverage matrix    │
│ • Web UI (:8080)     │                  │ • Web UI (:9090)     │
└──────────────────────┘                  └──────────────────────┘
          |                                         |
          ▼                                         ▼
     ┌──────────┐                            ┌──────────────┐
     │ ATT&CK   │                            │ Correlation  │
     │ Matrix   │                            │   & Gap      │
     │ Overlay  │                            │ Visibility   │
     └──────────┘                            └──────────────┘
```

Quick Start

Install

```
git clone https://github.com/ICSforge/ICSforge.git
cd ICSforge
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install -e .
chmod +x icsforge.sh
```

Web UI

```
sudo ./icsforge.sh web                                # Sender dashboard on :8080
sudo ./icsforge.sh receiver                           # Receiver dashboard on :9090
sudo ./icsforge.sh receiver --l2-iface eth0 # Receiver with Profinet Listener
```

Or with Docker

```
docker compose up
# Sender UI:   http://localhost:8080
# Receiver UI: http://localhost:9090
```

Protocol Coverage

Protocol	Port	Styles	Techniques (of 67)
OPC UA	TCP/4840	30	58/67
DNP3	TCP/20000	22	57/67
S7comm (Siemens)	TCP/102	36	56/67
EtherNet/IP (Allen-Bradley)	TCP/44818	23	55/67
Modbus/TCP	TCP/502	29	54/67
BACnet/IP	UDP/47808	16	54/67
MQTT	TCP/1883	16	52/67
IEC-104	TCP/2404	24	51/67
PROFINET DCP	L2/EtherType 0x8892	8	44/67
IEC 61850 GOOSE	L2/EtherType 0x88B8	5	42/67

IEC 61850 GOOSE and **PROFINET DCP** are Layer 2 protocols — they require a raw socket interface (`--l2-iface eth0`) for live sends. PCAP and offline generation work without a network interface.

Techniques at Full Coverage (10/10 protocols)

T0802, T0803, T0804, T0811, T0813, T0814, T0815, T0819, T0821, T0826, T0827, T0828, T0829, T0830, T0831, T0832, T0840, T0846, T0848, T0855, T0856, T0858, T0859, T0861, T0866, T0868, T0869, T0877, T0881, T0882, T0885, T0886, T0888, T0889, T0891 — **35 techniques** fully covered across all 10 protocols.

Scenarios

Scenarios are defined in `icsforge/scenarios/scenarios.yml`.

Naming convention: `T08XX__<description>__<protocol>[_variant]`

Types:

Type	Count	Description
Standalone scenarios	536	Single-technique, single-protocol runs
Named attack chains	11	Multi-step sequences modelling real campaigns

Named attack chains include:

- **Industroyer2** — Ukraine 2022 power grid attack (IEC-104 + S7comm)
- **SIS Targeting (TRITON-inspired)** — Safety system targeting (SIS), Modbus + S7comm surrogate
- **Stuxnet-style** — Siemens PLC programme manipulation
- **Water Treatment** — Oldsmar-style setpoint tampering (Modbus + IEC-104)
- **OPC UA Espionage** — Silent data exfiltration via OPC UA sessions
- **EtherNet/IP Manufacturing** — Allen-Bradley CIP manipulation
- **AitM + Sensor Spoof** — Adversary-in-the-Middle combined with DNP3 measurement injection
- **Firmware Persistence** — S7comm firmware implant and persistence
- **Full ICS Kill Chain** — Recon to impact (S7comm + Modbus)
- **Loss of Availability** — Multi-protocol concurrent DoS (DNP3 + IEC-104 + Modbus)
- **Default Creds → Impact** — Lateral pivot to programme modification

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What Techniques Are Covered

ICSForge implements 67 of 83 ATT&CK for ICS techniques at the network-observable level. The remaining 16 are correctly classified as host-only, require physical access, or generate no network-observable packets — they are documented in `icsforge/data/technique_support.json` with explicit rationale per technique, and also exposed via `/api/technique_support`.

ATT&CK matrix counts explained:

- 83 unique technique IDs in the matrix
- 94 total matrix entries — 11 techniques appear under multiple tactics (e.g. T0856 Spoof Reporting Message appears under both Evasion and Impair Process Control, which is correct ATT&CK for ICS design)

SENDER WEB UI

OT/ICS CYBERSECURITY COVERAGE VALIDATION PLATFORM

ATT&CK FOR ICS · EXECUTED → DELIVERED → DETECTED

OPEN SENDER →

ATT&CK MATRIX

ICSForge helps you to validate cybersecurity detection coverage across OT/ICS environments by generating realistic industrial network traffic aligned to ATT&CK for ICS v18. Fire live packets or generate PCAPs — then prove **Executed** → **Delivered** → **Detected** coverage across your OT environment.

Safe by design: destination validation, cooperative receiver, no destructive payloads.

536	68	10	4
SCENARIOS	TECHNIQUES	PROTOCOLS	INDUSTRY PROFILES

QUICK START

1. Start a receiver on a safe destination IP in your OT environment (or use loopback for a local demo).
2. Open Sender → pick a scenario or attack chain → set receiver destination → Run Live.
3. Use Campaign Playbooks to sequence multi-scenario attacks with dwell timers as a single run_id.
4. Open the ATT&CK Matrix to browse coverage — click any runnable tile to fire traffic and overlay results. Export a Coverage Report when done.

SCENARIO PACK

scenarios
547 scenarios · 10 protocols · 68 techniques

bacnet | dnp3 | enip | iec104 | iec61850 | modbus | mqtt | opcua | profinet_dcp | s7comm

TOP TECHNIQUES BY COVERAGE

ID	TECHNIQUE	PROTOCOLS	SCENARIOS
T0855	Unauthorized Command Message	10/10	38
T0889	Modify Program	10/10	38
T0848	Rogue Master	10/10	35
T0831	Manipulation of Control	10/10	31
T0888	Remote System Information Discovery	10/10	29
T0861	Point & Tag Identification	10/10	29

BROWSE SCENARIOS

Filter by ID, technique, or keyword...

SCENARIO ID	TITLE
CHAIN__aitm_sensor_spoof__opcua_dnp3	CHAIN – AitM + Sensor Spoofing via OPC UA + DNP3 (T0830+T0856+T0832)
CHAIN__default_creds_to_ics_impact__multi	CHAIN – Default Creds → Lateral Pivot → Program Modification → Impact
CHAIN__enip_manufacturing_attack	CHAIN – EtherNet/IP manufacturing line attack

OT/ICS CYBERSECURITY COVERAGE VALIDATION PLATFORM

ATT&CK FOR ICS · EXECUTED → DELIVERED → DETECTED

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SCENARIO ID	TITLE
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CHAIN__default_creds_to_ics_impact__multi	CHAIN – Default Creds → Lateral Pivot → Program Modification → Impact
CHAIN__enip_manufacturing_attack	CHAIN – EtherNet/IP manufacturing line attack

⚙ NETWORK SETTINGS

SENDER IP (SRC)	RECEIVER IP (DST)	RECEIVER PORT	SENDER CALLBACK URL (OPTIONAL OVERRIDE)	CALLBACK TOKEN (OPTIONAL)
192.168.178.10	192.168.178.30 → auto-fills Destination IP below	9090	http://sender-ip:8080/api/receiver/callback	shared-secret

○ PULL MODE SENDER-INITIATES | SAVE & CONNECT | ✓ Connected — callback active

⛓ ATTACK CHAINS — ONE-CLICK NAMED SCENARIOS

🏭 EtherNet/IP MFG
Allen-Bradley · CIP manipulation
T0836 T0840 T0868 T0881

🕵 OPC UA Espionage
Silent exfil via OPC UA sessions
T0801 T0822 T0861 T0879

⚡ Industroyer2
Ukraine 2022 · IEC-104 + S7comm
T0813 T0815 T0848 T0855

⚙ Stuxnet-style
Siemens PLC · Program manipulation
T0840 T0843 T0845 T0858

☢ SIS Targeting (TRITON-inspired)
Modbus+S7comm surrogate · SIS disable pattern
T0829 T0832 T0882 T0888

💧 Water Treatment
Oldsmar-style · Setpoint tampering
T0806 T0829 T0831 T0861

MULTI-PROTOCOL CHAINS

⛓ CHAIN – AitM + Sensor Spoofing via OPC UA + DNP3 (T0830+T0856+T0832)
dnp3 · opcua
T0830 T0832 T0856 T0882

⛓ CHAIN – Default Creds → Lateral Pivot → Program Modification → Impact
enip · iec104 · modbus · opcua · s7comm
T0812 T0831 T0855 T0886

⛓ CHAIN – Firmware Persistence via S7comm (T0889+T0839+T0857+T0858)
s7comm
T0839 T0857 T0858 T0888

⛓ CHAIN – Full ICS Kill Chain — Recon to Impact (S7comm-centric)
modbus · s7comm
T0812 T0813 T0835 T0838

⛓ CHAIN – Loss of Availability — Multi-protocol concurrent DoS (T0826+T0814+T0813)
dnp3 · iec104 · modbus · s7comm
T0813 T0814 T0826

SCENARIO

INDUSTRY PROFILE: All sectors

SEARCH: Filter by name, technique ID, or protocol...

INITIAL ACCESS

- T0822 – External Remote Services — EtherNet/IP v...
- T0822 – External Remote Services — Internet-expo...
- T0822 – External Remote Services — Internet-expo...
- T0822 – MQTT broker connection (External Remote ...
- T0822 – External Remote Services — S7comm via re...
- T0822 – OPC UA anonymous session establishment (...

Accesses a PLC or gateway exposed on TCP/502 through an internet connection. Modbus is the most commonly internet-exposed OT protocol — Shodan finds hundreds of thousands of accessible endpoints. With no authentication,

LIVE PAYLOAD PREVIEW

Actual PDU bytes from the real protocol builder — what hits the wire.

← PREV | Step 1 / 3 | NEXT →

modbus | style: read_holding | TCP :502 | 66 bytes | T0822

```
0000  f1 1f 00 00 00 3c cc 03 23 23 00 02 49 43 53 46   .....<..##..ICSF
0010  4f 52 47 45 3a 49 43 53 46 4f 52 47 45 3a 50 52   ORGE:ICSFORGE:PR
0020  45 56 49 45 57 3a 54 30 38 32 32 5f 5f 65 78 74   EVIEW:T0822__ext
0030  65 72 6e 61 6c 5f 72 65 6d 6f 3a 54 30 38 32 32   ernal_remo:T0822
0040  3a 3a                                             ::
```

SCENARIO SUMMARY

3 steps · 20 total packets modbus

INDUSTRY PROFILE

All sectors — show everything

Campaign Playbooks

Multi-scenario sequenced campaigns with configurable dwell timers between steps.

Full ICS Kill Chain

Comprehensive 9-scenario campaign covering the complete OT attack lifecycle: reconnaissance → creden...

9 scenarios | ~3 minutes

Stuxnet-Style Campaign

Reproduces the Stuxnet attack sequence: PROFINET device discovery, S7comm fingerprinting, program up...

5 scenarios | ~90 seconds

AitM + Sensor Spoofing

Adversary-in-the-Middle attack against OPC UA historian followed by spoofed sensor values injected i...

5 scenarios | ~75 seconds

OT Credential Harvest

Multi-protocol default credential probe across Modbus, S7comm, OPC UA, and EtherNet/IP followed by I...

5 scenarios | ~60 seconds

Safety System Attack (Triton-style)

Targets safety instrumented systems: reads SIS state, manipulates protection relays, disables safety...

5 scenarios | ~90 seconds

FULL ICS KILL CHAIN

Campaign Playbook

Comprehensive 9-scenario campaign covering the complete OT attack lifecycle: reconnaissance → credential access → collection → C2 → impact. Based on combined TTPs from Stuxnet, Industroyer, and Triton.

Destination IP: 192.168.178.30

Interface (PROFINET L2, optional): eth0

Steps

1	Phase 1 · Recon: EtherNet/IP device sweep	
2	Phase 1 · Recon: S7comm SZL fingerprint	+8s
3	Phase 2 · Initial Access: Default credentials	+10s
4	Phase 3 · Collection: Process state monitoring	+12s
5	Phase 3 · Collection: Full I/O image read	+10s
6	Phase 4 · C2: Covert beacon via Modbus registers	+15s
7	Phase 5 · Inhibit: Alarm threshold manipulation	+12s

RUN CAMPAIGN | ABORT | Running...

LIVE PROGRESS

run_id: 2026-04-10-X

✓	Phase 1 · Recon: EtherNet/IP device sweep	
✓	Phase 1 · Recon: S7comm SZL fingerprint	+8s
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4	Phase 3 · Collection: Process state monitoring	+12s
5	Phase 3 · Collection: Full I/O image read	+10s
6	Phase 4 · C2: Covert beacon via Modbus registers	+15s
7	Phase 5 · Inhibit: Alarm threshold manipulation	+12s
8	Phase 6 · Impact: PLC program modification	+15s
9	Phase 6 · Impact: CPU STOP + program reload	+10s

Event Stream

```
17:48:37 Step 2/9: Phase 1 · Recon: S7comm SZL fingerprint
17:48:55   ✓ 12 packet(s)
17:48:55   ⏱ waiting 10s…
17:48:56   ✓ 21 packet(s)
```

ATT&CK for ICS v18.1 — Interactive Matrix

RUNNABLE = live network traffic | PRECURSOR = observable approximation | – = host-only / not implemented

192.168.178.30 | SAVE IP | ALL | RUNNABLE | PRECURSOR | GAP

Overlay: 2026-04-10-X • Stuxnet-Style Campaign • 2026-04-10T15:50 | LOAD RUNS | APPLY | CLEAR

■ Executed ■ Detected ■ Gap

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
T0817 Drive-by Compromise	T0807 Command-Line Interface	T0839 Module Firmware	T0874 Hooking	T0820 Exploitation for Evasion	T0840 Network Connection Enumeration	T0812 Default Credentials	T0801 Monitor Process State	T0869 Standard Application Layer Protocol	T0800 Activate Firmware Update Mode	T0806 Brute Force I/O	T0813 Denial of Control
T0819 Exploit Public-Facing Application	T0821 Modify Controller Tasking	T0857 System Firmware	T0890 Exploitation for Privilege Escalation	T0849 Masquerading	T0842 Network Sniffing	T0843 Program Download	T0802 Automated Collection	T0884 Connection Proxy	T0803 Block Command Message	T0836 Modify Parameter	T0815 Denial of View
T0822 External Remote Services	T0823 Graphical User Interface	T0859 Valid Accounts		T0851 Rootkit	T0846 Remote System Discovery	T0859 Valid Accounts	T0811 Data from Information Repositories	T0885 Commonly Used Port	T0804 Block Reporting Message	T0839 Module Firmware	T0826 Loss of Availability
T0847 Replication Through Removable Media	T0834 Native API	T0873 Project File Infection		T0856 Spoof Reporting Message	T0887 Wireless Sniffing	T0866 Exploitation of Remote Services	T0830 Adversary-in-the-Middle		T0805 Block Serial COM	T0855 Unauthorized Command Message	T0827 Loss of Control
T0848 Rogue Master	T0853 Scripting	T0889 Modify Program		T0858 Change Operating Mode	T0888 Remote System Information Discovery	T0867 Lateral Tool Transfer	T0845 Program Upload		T0809 Data Destruction	T0856 Spoof Reporting Message	T0828 Loss of Productivity and Revenue
T0860 Wireless Compromise	T0858 Change Operating Mode	T0891 Hardcoded Credentials		T0872 Indicator Removal on Host		T0886 Remote Services	T0852 Screen Capture		T0814 Denial of Service		T0829 Loss of View
T0862 Supply Chain Compromise	T0863 User Execution			T0894 System Binary Proxy Execution		T0891 Hardcoded Credentials	T0861 Point & Tag Identification		T0816 Device Restart/Shutdown		T0831 Manipulation of Control
T0864 Transient Cyber Asset	T0871 Execution through API						T0868 Detect Operating Mode		T0835 Manipulate I/O Image		T0832 Manipulation of View
T0865 Spearphishing Attachment	T0874 Hooking						T0877 I/O Image		T0838 Modify Alarm Settings		T0837 Loss of Protection
T0866 Exploitation of Remote Services	T0895 Autorun Image						T0887 Wireless Sniffing		T0851 Rootkit		T0879 Damage to Property
T0883 Internet Accessible Device							T0893 Data from Local System		T0857 System Firmware		T0880 Loss of Safety
T0886 Remote Services									T0878 Alarm Suppression		T0882 Theft of Operational Information
									T0881 Service Stop		
									T0892 Change Credential		

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T0822 External Remote Services	T0823 Graphical User Interface	T0859 Valid Accounts		T0851 Rootkit	T0846 Remote System Discovery	T0859 Valid Accounts	T0811 Data from Information Repositories	T0885 Commonly Used Port	T0804 Block Reporting Message	T0839 Module Firmware	T0826 Loss of Availability
T0847 Replication Through Removable Media	T0834 Native API	T0873 Project File Infection		T0856 Spoof Reporting Message	T0887 Wireless Sniffing	T0866 Exploitation of Remote Services	T0830 Adversary-in-the-Middle		T0805 Block Serial COM	T0855 Unauthorized Command Message	T0827 Loss of Control
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T0860 Wireless Compromise	T0858 Change Operating Mode	T0891 Hardcoded Credentials		T0872 Indicator Removal on Host		T0886 Remote Services	T0852 Screen Capture		T0814 Denial of Service		T0829 Loss of View
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T0866 Exploitation of Remote Services	T0895 Autorun Image						T0887 Wireless Sniffing		T0851 Rootkit		T0879 Damage to Property
T0883 Internet Accessible Device							T0893 Data from Local System		T0857 System Firmware		T0880 Loss of Safety
T0886 Remote Services									T0878 Alarm Suppression		T0882 Theft of Operational Information
									T0881 Service Stop		
									T0892 Change Credential		

Overlay applied
run_id: 2026-04-10-X

COVERAGE REPORT

Build an ATT&CK for ICS heatmap. Fill in what was tested, what was detected, and where gaps were found. Export as a self-contained HTML report.

Run Reference

Select a previous run

★ All runs combined (50 runs) | LOAD

or paste run ID manually

Loaded 38 unique technique(s) across 50 runs

Technique Coverage

Executed — fired by ICSForge

T0801
T0805
T0806
T0807
T0812

Detected — confirmed by NSM / SIEM

T0877
T0878
T0886
T0888
T0889

Gaps Found — executed but not detected (blind spots)

T0879
T0881
T0882
T0826
T0828

Protocol Coverage Gaps

Flag specific protocols that are not firewalled, monitored, or have known blind spots. These appear as a dedicated table in the report.

Modbus/TCP (502 | Not firewalled | ×

ICSForge Coverage Report

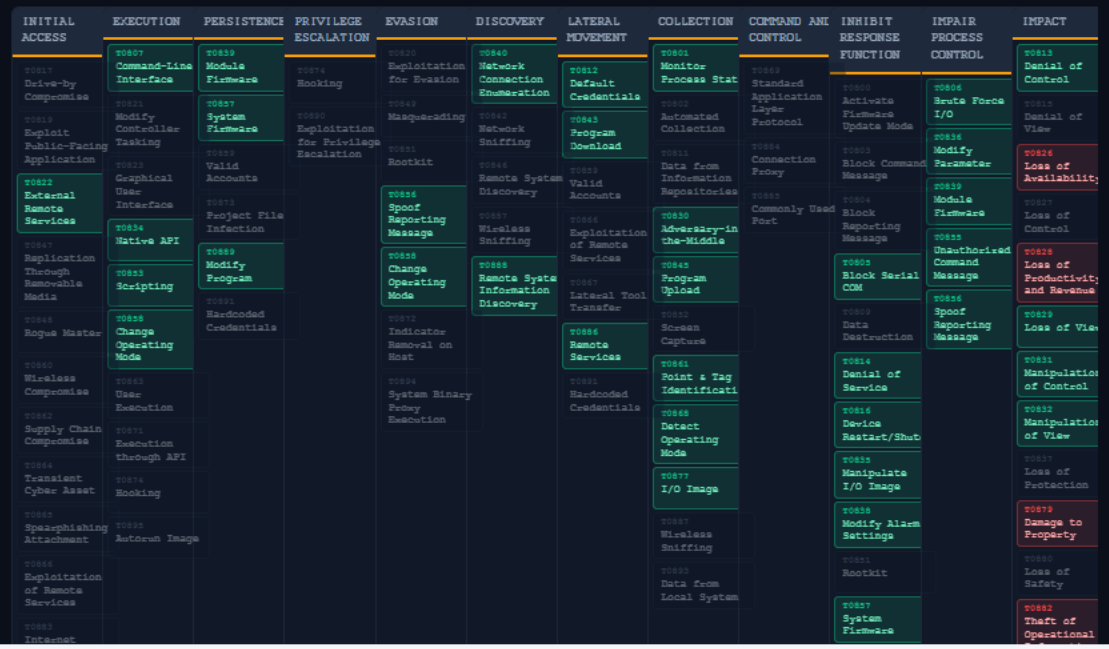
Assessment Date: 2026-04-10 · **Generated:** 2026-04-10 15:53 UTC
Run ID: — · **Matrix:** ATT&CK for ICS v18.1
Organization: Test Co. · **Analyst:** John Doe
Notes: 1st Test

94	38	33	5	40%
MATRIX TECHNIQUES	EXECUTED	DETECTED	GAPS FOUND	MATRIX COVERAGE

86%	3
DETECTION RATE	PROTOCOL GAPS

ATT&CK FOR ICS HEATMAP

Detected · Executed / Not Detected · Gap Found (blind spot) · Not Tested



TOOLS

Convenience UI for key workflows. Sender mode includes offline artifacts and replay. Receiver mode is read-only and minimal.

HEALTH & SELF-TEST

Quick sanity checks for demo readiness.

HEALTH

OUTPUT

```
{
  "capabilities": {
    "af_packet": true,
    "pcap_write": true,
    "profinet_live": true,
```

INTERFACES

Discover local network interfaces (useful for replay and live send).

REFRESH

```
eth0
wlan0
```

OFFLINE GENERATE

Generate ground-truth JSONL and optional PCAP from `icsforge/scenarios/scenarios.yml` without live sending.

SCENARIO

filter scenarios (e.g. T0855 / modbus)

T0814__denial_of_service__dnp3_flood

Choose a scenario

OUTDIR: out

DST IP (FOR ARTIFACTS): 192.168.178.30

SRC IP (FOR ARTIFACTS): 192.168.178.10

✓ BUILD PCAP GENERATE

⬇ DOWNLOAD PCAP — T0814__DENIAL_OF_SERVICE__2026-04-10__QUEBEC.PCAP

RESULT

```
✓ Events: /home/kali/ICSforge/out/events/T0814__denial_of_service__2026-04-10__QUEBEC.jsonl
✓ PCAP:   /home/kali/ICSforge/out/pcaps/T0814__denial_of_service__2026-04-10__QUEBEC.pcap
```

PCAP REPLAY

Replay a PCAP toward a destination IP (requires sudo/root for raw send on many systems).

PCAP PATH (SERVER-SIDE) OR UPLOAD BELOW

/home/kali/ICSforge/out/pcaps/uploads/T0881__service_stop__2026-04-10__ZULU.pcap

Upload PCAP: Choose File UPLOAD ✓ T0881__service_stop__2026-04-10__ZULU.pcap

REPLAY

RESULT

```
Working ...
```

RECEIVER WEB UI

RECEIVER

Passive proof-of-delivery sink · no responses · no PLC writes

Last: 2026-04-10 15:52:34 PASSIVE ↻ RESET

1918	42	8	8
PACKETS RECEIVED	ACTIVE RUNS	TECHNIQUES SEEN	PROTOCOLS SEEN

○ PROFINET L2 listener active on interface: eth0

SENDER CALLBACK SETTINGS

Configure where this receiver should send live receipt confirmations.

SENDER CALLBACK URL: http://sender-ip:8080/api/receiver/callback

CALLBACK TOKEN (OPTIONAL): shared-secret

SAVE CALLBACK | TEST CALLBACK

LIVE RECEIPTS

RUN ID: e.g. 2026-03-05-BRAVO-07 | TECHNIQUE: e.g. T0855 | PROTOCOL: e.g. modbus | APPLY

TIME	PROTO	SOURCE	TECHNIQUE	RUN ID	BYTES
2026-04-10 15:52:34	s7comm	192.168.178.10:50318	T0843	2026-04-10 NOVEMBER-71	130
2026-04-10 15:52:34	s7comm	192.168.178.10:50304	T0843	2026-04-10 NOVEMBER-71	130
2026-04-10 15:52:34	s7comm	192.168.178.10:50300	T0843	2026-04-10 NOVEMBER-71	117
2026-04-10 15:52:33	s7comm	192.168.178.10:50296	T0843	2026-04-10 NOVEMBER-71	117
2026-04-10 15:52:33	s7comm	192.168.178.10:50280	T0843	2026-04-10 NOVEMBER-71	332
2026-04-10 15:52:33	s7comm	192.168.178.10:50274	T0843	2026-04-10 NOVEMBER-71	330

RUN DETAILS

Click a run_id in the table to inspect.

2026-04-10-NOVEMBER-71 DELIVERED

Packets: 33
First: 2026-04-10 15:52:31
Last: 2026-04-10 15:52:34

Techniques observed:

T0843

Evidence stored in receipts.jsonl with correlation marker.

RECEIVER

Passive proof-of-delivery sink · no responses · no PLC writes

Last: 2026-04-10 15:52:34 PASSIVE RESET

1918	42	8	8
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LIVE RECEIPTS

RUN ID: e.g. 2026-03-05-BRAVO-07 | TECHNIQUE: e.g. T0855 | PROTOCOL: e.g. modbus | APPLY

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2026-04-10 15:52:34	s7comm	192.168.178.10:50300	T0843	2026-04-10 NOVEMBER-71	117
2026-04-10 15:52:33	s7comm	192.168.178.10:50296	T0843	2026-04-10 NOVEMBER-71	117
2026-04-10 15:52:33	s7comm	192.168.178.10:50280	T0843	2026-04-10 NOVEMBER-71	332
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RUN DETAILS

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ATT&CK for ICS v18.1 — Interactive Matrix

RUNNABLE = live network traffic PRECURSOR = observable approximation – = host-only / not implemented

ALL | RUNNABLE | PRECURSOR | GAP

Overlay: (no overlay) LOAD RUNS APPLY CLEAR

Executed | Detected | Gap

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
T0817 Drive-by Compromise	T0807 Command-Line Interface	T0839 Module Firmware	T0874 Hooking	T0820 Exploitation for Evasion	T0840 Network Connection Enumeration	T0812 Default Credentials	T0801 Monitor Process State	T0869 Standard Application Layer Protocol	T0800 Activate Firmware Update Mode	T0806 Brute Force I/O	T0813 Denial of Control
T0819 Exploit Public-Facing Application	T0821 Modify Controller Tasking	T0857 System Firmware	T0890 Exploitation for Privilege Escalation	T0849 Masquerading	T0842 Network Sniffing	T0843 Program Download	T0802 Automated Collection	T0884 Connection Proxy	T0803 Block Command Message	T0836 Modify Parameter	T0815 Denial of View
T0822 External Remote Services	T0823 Graphical User Interface	T0859 Valid Accounts		T0851 Rootkit	T0846 Remote System Discovery	T0859 Valid Accounts	T0811 Data from Information Repositories	T0885 Commonly Used Port	T0804 Block Reporting Message	T0839 Module Firmware	T0826 Loss of Availability
T0847 Replication Through Removable Media	T0834 Native API	T0873 Project File Infection		T0856 Spoof Reporting Message	T0887 Wireless Sniffing	T0866 Exploitation of Remote Services	T0830 Adversary-in-the-Middle		T0805 Block Serial COM	T0855 Unauthorized Command Message	T0827 Loss of Control
T0848 Rogue Master	T0853 Scripting	T0889 Modify Program		T0858 Change Operating Mode	T0888 Remote System Information Discovery	T0867 Lateral Tool Transfer	T0845 Program Upload		T0809 Data Destruction	T0856 Spoof Reporting Message	T0828 Loss of Productivity and Revenue
T0860 Wireless Compromise	T0858 Change Operating Mode	T0891 Hardcoded Credentials		T0872 Indicator Removal on Host		T0886 Remote Services	T0852 Screen Capture		T0814 Denial of Service		T0829 Loss of View
T0862 Supply Chain Compromise	T0863 User Execution			T0894 System Binary Proxy Execution		T0891 Hardcoded Credentials	T0861 Point & Tag Identification		T0816 Device Restart/Shutdown		T0831 Manipulation of Control
T0864 Transient Cyber Asset	T0871 Execution through API						T0868 Detect Operating Mode		T0835 Manipulate I/O Image		T0832 Manipulation of View
T0865 Spearphishing Attachment	T0874 Hooking						T0877 I/O Image		T0838 Modify Alarm Settings		T0837 Loss of Protection
T0866 Exploitation of Remote Services	T0895 Autorun Image						T0887 Wireless Sniffing		T0851 Rootkit		T0879 Damage to Property
T0883 Internet Accessible Device							T0893 Data from Local System		T0857 System Firmware		T0880 Loss of Safety
T0886 Remote Services									T0878 Alarm Suppression		T0882 Theft of Operational Information
									T0881 Service Stop		

TOOLS

Convenience UI for key workflows. Sender mode includes offline artifacts and replay. Receiver mode is read-only and minimal.

HEALTH & SELF-TEST

Quick sanity checks for demo readiness.

HEALTH

OUTPUT


```
{
  "capabilities": {
    "af_packet": true,
    "pcap_write": true,
    "profinet_live": true,
    "tcp_send": true
  },
  "ok": true,
  "receiver": null,
  "receiver_error": null,
  "receiver_ok": null,
  "repo_root": "/home/kali/ICSforge",
  "scenario_pack_count": 1,
  "scenario_packs": [
    "/home/kali/ICSforge/icsforge/scenarios/scenarios.yml"
  ],
  "timestamp": "2026-04-10T16:00:52.350215+00:00Z",
  "version": "0.59.4"
}
```


INTERFACES

Discover local network interfaces (useful for replay and live send).

REFRESH

eth0
wlan0

EXAMPLE DETECTIONS IN OT NSM SENSOR

Home > Alerts

Defender for IoT | Alerts

Refresh | Edit Columns | Export to CSV | Change Status | Export to PDF | Learn

Search | Status == **New** | Last detection == **Last day** | Source Device == **192.168.178.10** | Destination Device == **192.168.178.30** | Add filter | Reset filters

Showing 324 of 616 alerts

Group by: Severity

Severity	Name ↓	Engine	Last detection	Status	Source Device	Destination Device
Critical (1)						
Critical	Unauthorized PLC Programming	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major (321)						
Major	Unpermitted Usage of Modbus Function Code	Policy Violation	4 hours ago	New	192.168.178.10	192.168.178.30
Major	Unpermitted Usage of ASDU Types	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	Unauthorized Siemens S7 Execution of User Defir	Policy Violation	4 hours ago	New	192.168.178.10	192.168.178.30
Major	Unauthorized Siemens S7 Execution of Control Fu	Policy Violation	4 hours ago	New	192.168.178.10	192.168.178.30
Major	Unauthorized PLC Program Upload	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	Unauthorized HTTP Activity	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	Unauthorized BACNet Route	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	Unauthorized BACNet Object Access	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	New Activity Detected - EtherNet/IP Protocol Cor	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	MQIsdp New Topic Subscription Detected	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	MQIsdp New Publish Detected	Policy Violation	5 hours ago	New	192.168.178.10	192.168.178.30
Major	Modbus Address Range Violation	Policy Violation	4 hours ago	New	192.168.178.10	192.168.178.30
Major	Modbus Address Range Violation	Policy Violation	4 hours ago	New	192.168.178.10	192.168.178.30

https://www.icsforge.nl